ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

MATHEMATICS

PAPER 1

MARKS: 100 **TIME: 2 hours**

This question paper consists of 6 pages.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Clearly show ALL calculations, diagrams, graphs, et cetera that you have used to determine your answers.
3. Answers only will NOT necessarily be awarded full marks.
4. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
5. If necessary, round off answers to TWO decimal places, unless stated otherwise.
6. Diagrams are NOT necessarily drawn to scale.
7. Number the answers correctly according to the numbering system used in this question paper.
8. Write neatly and legibly.

QUESTION 1

1.1 Simplify fully:

1.1.1 $(2x-3)(x^2+4x-1)$ (3)

1.1.2 $\dfrac{x^2-9}{x^2+x-6} \div \dfrac{x-3}{x-2}$ (4)

1.2 Factorise fully:

1.2.1 $3x^2-12$ (2)

1.2.2 $2x^2-7x-15$ (2)

1.2.3 x^3-8 (2)

1.2.4 $ax-ay+2x-2y$ (2)

1.3 Simplify WITHOUT using a calculator:

1.3.1 $\dfrac{2^{x+2}-2^x}{3\cdot 2^x}$ (3)

1.3.2 $27^{2/3}\times 4^{-1/2}$ (3)

1.3.3 $(x^3y^{-2})^2\times x^{-4}y^5$ (3)

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QUESTION 2

2.1 Solve for x:

2.1.1 $3(x-2)=2x+5$ (2)

2.1.2 $x^2-5x-14=0$ (3)

2.1.3 $\frac{x+1}{2}-\frac{x-3}{3}=2$ (3)

2.1.4 $3^{x+1}=81$ (2)

2.1.5 $x(x-3)=10$ (3)

2.2 Solve for x and represent your answer on a number line: $-4<2x-2\leq 6$ (3)

2.3 Solve for x and y simultaneously: (4)
$x+2y=7$ and $3x-y=7$

2.4 Make i the subject of the formula: $A=P(1+in)$ (2)

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QUESTION 3

Consider the linear number pattern: $5\ ;\ 9\ ;\ 13\ ;\ 17\ ;\ ...$

3.1 Write down the next TWO terms of the pattern. (2)

3.2 Determine a formula for the general term, T_n. (2)

3.3 Calculate the value of T_{50}. (2)

3.4 Which term of the pattern is equal to 157? (2)

3.5 Show that 100 is NOT a term of this pattern. (2)

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QUESTION 4

Given: $f(x) = x^2 - 4$ and $g(x) = -2x + 4$

4.1 Write down the coordinates of the y-intercept of f. (1)

4.2 Calculate the coordinates of the x-intercepts of f. (2)

4.3 Calculate the coordinates of the x-intercept of g. (1)

4.4 Sketch the graphs of f and g on the same set of axes. Clearly show all intercepts with the axes and the turning point of f. (5)

4.5 Write down the range of f. (1)

4.6 Determine the values of x for which $f(x) \geq g(x)$. (4)

4.7 The graph of $h(x) = \frac{a}{x} + q$ has a horizontal asymptote $y = 1$ and passes through the point $(2\ ;\ 4)$. Determine the values of a and q. (3)

4.8 For $k(x) = 2^x$, write down:

4.8.1 The equation of the asymptote of k (1)

4.8.2 The coordinates of the y-intercept of k (1)

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Use the grid below for QUESTION 4.4, or draw the sketch in your answer book.



QUESTION 5

5.1 Thandi invests R8 000 for 5 years at a **simple** interest rate of 7% per annum. Calculate the amount Thandi will have at the end of 5 years. (3)

5.2 Sipho invests R8 000 for 5 years at an interest rate of 6,5% per annum, **compounded annually**. Calculate the amount Sipho will have at the end of 5 years. (3)

5.3 Who will have more money after 5 years, and by how much? (2)

5.4 A television costs R6 500 today. If the average inflation rate is 5% per annum, calculate how much the same television will cost in 3 years' time. (3)

5.5 The exchange rate is \$1 = R18,40. How many US dollars can you buy with R2 300? (2)

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QUESTION 6

In a class of 40 learners, 22 learners play soccer (S), 18 learners play netball (N) and 7 learners play both soccer and netball.

6.1 Draw a Venn diagram to represent this information. (4)

6.2 A learner is chosen at random from the class. Determine the probability that the learner:

6.2.1 Plays neither soccer nor netball (2)

6.2.2 Plays soccer or netball (2)

6.2.3 Plays only soccer (2)

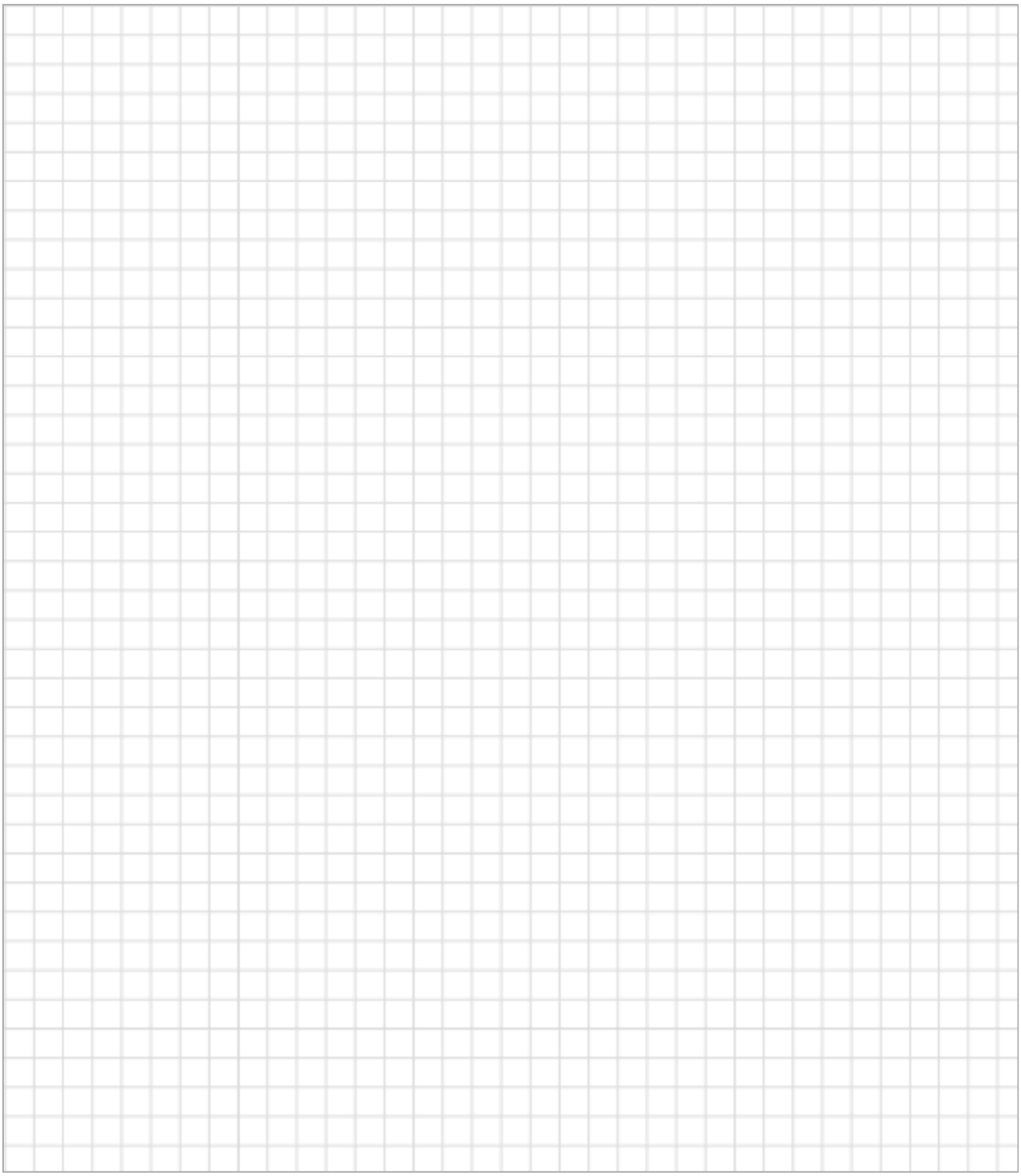
6.3 Are the events S and N mutually exclusive? Give a reason for your answer. (2)

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TOTAL: 100

SPACE FOR ROUGH WORK

This page will not be marked.



END OF QUESTION PAPER